

A Charged Particle on a Flat Magnetic Torus: Clean Cyclotron Return and Sharp Flux Quantization

HCS-C376 theorem package

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Abstract

We close, in one normalization, the classical and quantum dynamics of a charged unit-mass particle on a rectangular flat two-torus with nonzero constant magnetic field. Every positive-energy classical trajectory has least period $2\pi/|B|$; each return is maximally clean, whereas no other time fixes a point of an energy shell. The cyclotron center identifies the orbit quotient as a two-torus. A quantum line bundle with curvature $-iB \, dx \wedge dy$ exists precisely at integral flux $N = BL_x L_y / (2\pi)$. Necessity follows from Chern–Weil theory and sufficiency from an explicit rectangle cocycle. This first round closes the global classical return and topological existence questions only.

中文摘要

本文在同一套符号和物理单位中研究矩形平坦二环面上的恒定磁场运动。经典部分不从局部轨道猜测周期，而是在普适覆盖上直接积分方程：动量作匀速旋转，回旋中心保持不变，位置由同一旋转精确重建。由此证明任意正能量轨道的最小相空间周期均为二派除以磁场绝对值；非返回时刻没有不动点，返回时刻整个能量层连同其切空间同时固定，因此所得周期族是最大干净连续族，其轨道商仍为二环面。量子存在性则由全局拓扑决定。对曲率负虚数乘以磁场面积形式的厄米线丛，第一陈数恰为磁通除以二派；只有该数为整数时对象存在。必要性来自陈类积分，充分性由基本矩形上的显式过渡函数给出。本轮只封闭经典回归与磁通量子化，不预告后续谱结论。

Keywords: magnetic torus; cyclotron flow; clean return; flux quantization; line bundle
关键词：磁环面；回旋流；干净回归；磁通量子化；线丛

1 Classical magnetic flow

Let

$$\mathbb{T}^2 = \mathbb{R}^2 / (L_x \mathbb{Z} \oplus L_y \mathbb{Z}), \quad A = L_x L_y,$$

where $L_x, L_y > 0$. On $T^*\mathbb{T}^2$ use lifted coordinates $(q, p) = (x, y, p_x, p_y)$ and the twisted form and Hamiltonian

$$\omega_B = dx \wedge dp_x + dy \wedge dp_y + B \, dx \wedge dy, \quad H_{\text{cl}} = \frac{1}{2}|p|^2. \quad (1)$$

Fix $B \neq 0$ and write $J(u, v) = (-v, u)$. Our Hamiltonian convention gives

$$\dot{q} = p, \quad \dot{p} = BJp. \quad (2)$$

Theorem 1 (Complete return and clean fixed sets). *On every energy shell $\Sigma_E = \{H_{\text{cl}} = E\}$ with $E > 0$, every orbit has the same least period*

$$T_{\text{cl}} = \frac{2\pi}{|B|}.$$

For $t \notin T_{\text{cl}}\mathbb{Z}$, $\text{Fix}(\Phi_t|_{\Sigma_E})$ is empty. For $t \in T_{\text{cl}}\mathbb{Z}$, the return is the identity and

$$\text{Fix}(\Phi_t|_{\Sigma_E}) = \Sigma_E, \quad \ker(D\Phi_t - I) = T\Sigma_E.$$

Thus every nonzero return is maximally clean. The orbit space of the circle action on $\Sigma_E \cong \mathbb{T}^2 \times S^1$ is a two-torus.

Proof. Let $R_\theta = \exp(\theta J)$. Direct integration of (2) gives

$$p(t) = R_{Bt} p_0, \quad c = q + \frac{Jp}{B} = q_0 + \frac{Jp_0}{B}, \quad q(t) = c - \frac{JR_{Bt} p_0}{B}. \quad (3)$$

This formula is well defined after reducing q modulo the lattice and proves completeness. Since $E > 0$, $p_0 \neq 0$. A phase-space return forces $R_{Bt}p_0 = p_0$, hence $Bt \in 2\pi\mathbb{Z}$; the position then returns as well. This proves both the period and its minimality. At an integral multiple of T_{cl} , formula (3) is the identity for all initial data, so its derivative is the identity. At every other time its rotation has no nonzero fixed momentum. These two observations give the stated fixed sets and clean tangent equality. Finally c modulo the lattice labels an orbit, proving the quotient assertion. $\square$

Geometric meaning. The continuous center c is the classical accidental symmetry. Compact configuration does not isolate trajectories: at a return time the entire three-dimensional energy shell is fixed. This fact is decisive for the Route-A verdict below.

2 Flux integrality and the quantum object

We use units $\hbar = q_{\text{charge}} = m = 1$. A quantum magnetic field means a Hermitian line bundle $\mathcal{L} \rightarrow \mathbb{T}^2$ with unitary connection ∇ whose curvature is

$$F_{\nabla} = -iB \, dx \wedge dy. \quad (4)$$

Theorem 2 (Sharp flux condition). *A pair $(\mathcal{L}, \nabla)$ satisfying (4) exists if and only if*

$$N = \frac{BA}{2\pi} \in \mathbb{Z}. \quad (5)$$

Its degree is N . For a fixed N , tensoring by a flat line bundle gives the two holonomy parameters and does not change the curvature.

Proof. Chern–Weil theory gives

$$c_1(\mathcal{L})[\mathbb{T}^2] = \frac{i}{2\pi} \int_{\mathbb{T}^2} F_{\nabla} = \frac{BA}{2\pi},$$

which proves necessity. Conversely, for $N \in \mathbb{Z}$ take on the fundamental rectangle the gauge $A_y = Bx$, $A_x = 0$. Identify the vertical sides trivially and identify $x = L_x$ with $x = 0$ by the unitary multiplier $\exp(iBL_x y)$. Its change around the horizontal cycle is $\exp(iBA) = \exp(2\pi iN) = 1$, so the corner cocycle closes and defines the required bundle and connection. Multiplying the two gluing maps by constant phases produces all flat holonomy twists. $\square$

Round-zero certificate: round zero proves complete clean classical return and the sharp global flux condition.